
UNIT 1 DATA SCIENCE FUNDAMENTALS

Structure

- 1.0 Introduction
- 1.1 Objectives
- 1.2 Understanding Data Science
- 1.3 Data Science Process
 - 1.3.1 Goal Identification
 - 1.3.2 Data Collection
 - 1.3.3 Data Preparation
 - 1.3.4 Exploratory Data Analysis (EDA)
 - 1.3.5 Model Development
 - 1.3.6 Model Deployment and Presentation
- 1.4 Data Science Vs Data Analytics
- 1.5 Types of Data Analysis
 - 1.5.1 Descriptive Analysis
 - 1.5.2 Diagnostic Analysis
 - 1.5.3 Predictive Analysis
 - 1.5.4 Prescriptive Analysis
- 1.6 Philosophy of Exploratory Data Analysis
- 1.7 Role of Data Scientist and Data Analyst
- 1.8 Crowd Sourcing
- 1.9 Languages for Data Science
- 1.10 Applications of Data Science
- 1.11 Data science Life Cycle
- 1.12 Summary
- 1.13 Solutions/Answers
- 1.14 Further Readings

1.0 INTRODUCTION

We can't escape data. Data can be generated each time you walk, such as step count; each time you eat, such as protein intake; each time you sleep, such as daily sleep duration; or each time you use a machine, such as energy consumption. When data is acquired and utilised effectively, it may enhance business prospects to augment profit. Due to the pervasive nature of data and its potential in improving businesses, a systematic method for acquiring, understanding and extracting key business insights are needed. This is where data science comes into play. This unit will introduce you to the fundamentals of data science.

Data science helps you to see things through an entirely new lens. Walking becomes more appealing when you can track your steps and use this

information to predict reduction in risk for cardiovascular events or drop in your annual health insurance premium. Isn't it fascinating that your social media handle displays advertisements, offers and posts of your interest. Such personalized marketing is changing the way we buy products. Cars may now notify users of imminent faults by using data from sensors embedded in them, giving an entirely new dimension to the use and maintenance of cars. CCTV (closed-circuit television) monitoring is no longer passive surveillance; instead, by utilising video analytics and Artificial Intelligence (AI), you may actively respond to a specific threat or undesirable condition without a security person physically sitting in front of the monitor. Government can use data science to make policies more effective. In this unit we will highlight many other applications of data science.

The aforementioned examples illustrate the immense potential of data science; yet, in order to be successful in this field, one must possess specialised knowledge that includes mathematics and statistics, analytics, Artificial Intelligence (AI), Machine Learning (ML) and computer programming. Additionally, appropriate domain knowledge is essential for identifying actionable insights that are concealed within an organization's data and utilising them for strategic planning and decision-making. There is a closely related field known as data analytics. This unit will provide you with key differences between these two closely related terms and highlight the roles of data scientist and data analytics in any organization. This unit will also take you through various types of data analysis techniques used in data science.

The steps in the data science life cycle include identifying a business problem, establishing the analysis's objectives, gathering, preparing, exploring, and modelling data, as well as deploying and presenting the outcomes. These stages will be covered in detail in this unit.

A significant volume of diverse data types is frequently required for the analysis of any data science endeavour, necessitating the use of a computer program. This unit will furnish you with an overview of several programming languages and software tools that are frequently employed in data science.

1.1 OBJECTIVES

After completing this unit, you should be able to:

- Understand process of data science
- Understand various applications of data science
- Learn key differences between data science and data analytics

1.2 UNDERSTANDING DATA SCIENCE

“Data science is the application of tools, methods, and techniques, including programming, statistics, machine learning, and algorithms, to integrate and analyse extensive databases for the purpose of deriving insights.”

The datasets are frequently composed of both structured and unstructured data. To comprehend this definition let us take an example. Imagine a critical

care unit in a multi-specialty hospital where patients are normally put on mechanical ventilation. Mechanical ventilation while lifesaving, also incurs risk of Ventilator-Induced Lung Injury (VILI). Optimal mechanical ventilation and weaning strategy is required. Where weaning is a process that gradually reduces the amount of support from the mechanical ventilator to allow the patient to breathe on their own. Experts' decisions might be subjective and may not always be optimal. Data science may support the experts in improved detection of patient conditions enabling timely intervention and personalized treatment which ultimately improves patient outcome and optimizes resources utilization in critical care. This provides a goal (*purpose*) for a data science project for this case. Many times, such goals are directly provided by businesses (hospitals or healthcare experts). However, in most cases, data scientists are required to work closely with domain experts (doctors and other healthcare professionals) to define the problem and specify goals. Domain experts also help in understanding various sources of data, available data, understanding physics of the underlying process (in this case mechanical ventilation and weaning) and challenges in capturing/recording new data. In this specific case, there could be multiple challenges such as data privacy, compliance involved in installing new sensors and required accuracy of results. Data scientists need to capture all such aspects while solving the problem at hand.

Data from multiple sources such as ventilator, para monitor, laboratory tests, radiology tests, electronic health records, and nursing charts are required. *Huge datasets* generated from these sources are also diverse in nature. Some of the data are monitored continuously, while some are monitored with lesser frequency such as once a week, thrice a week or every hour. Some are structured data like temperature, blood pressure, oxygen saturation, etc. Some are unstructured data such as X-ray and CT scan images, reports and handwritten clinical notes. Some are meta data such as height, registration number, etc. Some data are manually recorded, while some are extracted from sensors in digital form.

Various *tools and programming languages and libraries* like SQL, Python libraries (e.g., pandas, NumPy), R etc., are required for storing, manipulating and retrieving data in databases and data analysis. Data are normally visualized initially to identify patterns, trends, and anomalies before proceeding for further analytics. For example, in this case data scientists may try to find if blood pressure is showing any relationship with VILI. Regression, decision trees, clustering, artificial neural networks and other techniques are used to model data for prediction, classification or clustering. Tools such as Scikit-learn, TensorFlow, and PyTorch are normally used by data scientists. The outcome in this case would be prediction of health conditions to support weaning strategy. Before deployment it is important to assess model performance using metrics such as accuracy, precision and recall. The outcome of the model is then presented in the form of actionable insights. For example, insights in terms of reduction in risk of VILI, or early detection of VILI, etc. along with any mitigation plan can be presented in consultation with domain experts. Once validated, the proposed model needs to be integrated into the system for real world use. Tools such as Flask, Django, AWS and Docker are normally used for the deployment purpose.

Figure 1.1 summarizes the understanding of data science discussed above.

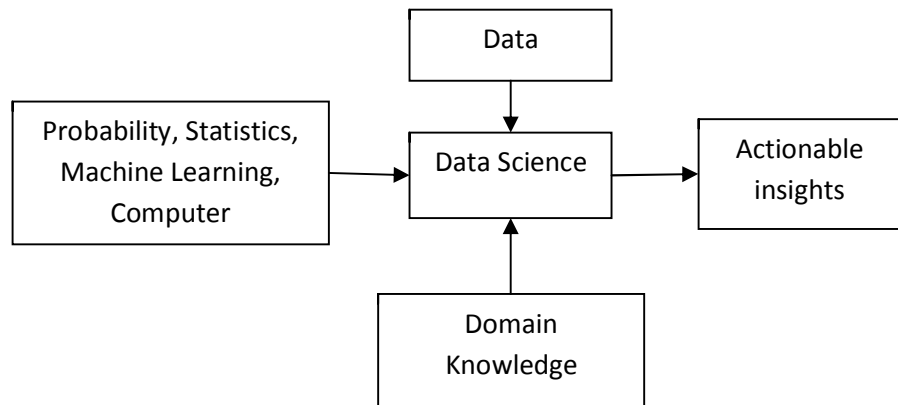


Figure 1.1: Functional blocks of data science

☞ Check Your Progress 1

1. What are key functional blocks of data science.

.....

.....

.....

2. What types of data are involved in critical care datasets and how are they categorized?

.....

.....

.....

3. How can data science support optimal mechanical ventilation and weaning in critical care?

.....

.....

.....

4. What are the main tools and techniques used by data scientists for analyzing healthcare data in the given case?

.....

.....

.....

5. Why is collaboration with domain experts important in a data science project related to critical care?

.....

.....

.....

1.3 DATA SCIENCE PROCESS

Data science is a systematic and iterative process. It involves several steps. When new insights or issues arise, data scientists frequently iterate back to previous steps. The data science process involves the following steps:

Step 1: Goal identification

Step 2: Data collection

Step 3: Data preparation

Step 4: Exploratory Data Analysis (EDA)

Step 5: Model development

Step 6: Model deployment and presentation

The steps are detailed below.

1.3.1 Goal Identification

This is the first stage in any data science project. It aims to understand the business problem and determine the objectives of the investigation. This demands a grasp of the project's 'what' and 'why'.

- What does management expect you to do?
- Why does management want this?

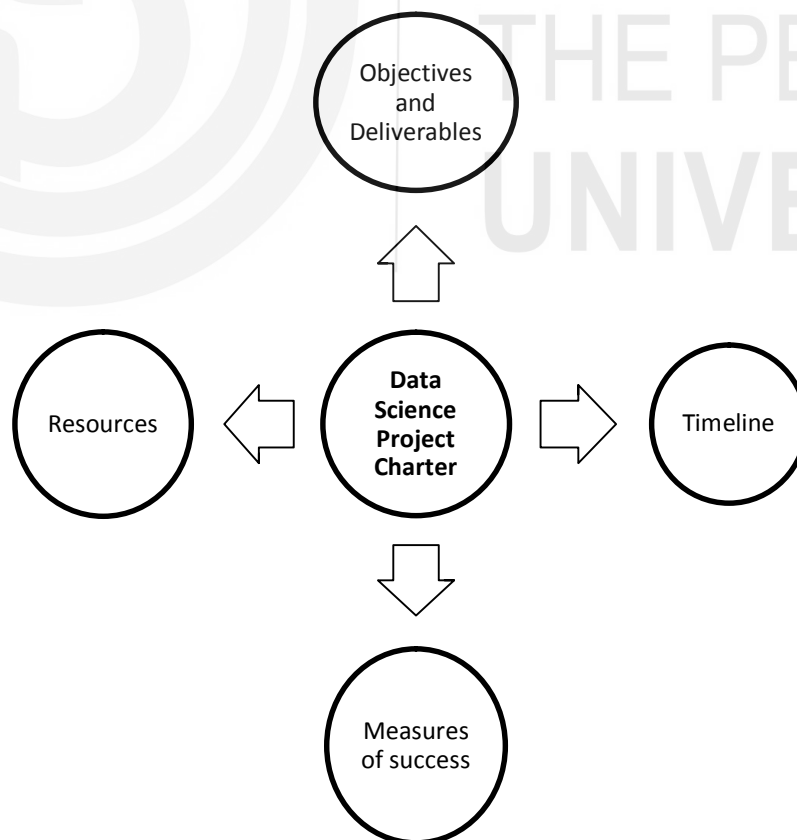


Figure 1.2: Constituents of a data science project charter

To successfully complete this stage, it is essential to ask questions until you have a comprehensive understanding of the business expectations, the project's role in the broader context, the project's impact on the business, the intended use of the project's outcomes, and the challenges or constraints. Upon completion of this phase, you can generate a project charter. The primary components of a data science project charter are illustrated in Figure 1.2. This step will guarantee that all stakeholders are aligned and that the project does not fail owing to a failure to capture the appropriate business requirements and context. Furthermore, this will assist management in estimating project expenses and organising necessary data and human resources. At the end of this step, you will have a clear understanding of the business problem.

Furthermore, it is prudent to build a proof of concept at this juncture to demonstrate the feasibility of the proposed project. It is evident that data scientists must possess business acumen, interpersonal skills, and effective communication to properly complete this phase.

Example: To further comprehend this step, consider the following example.

A company selling its products online via an e-commerce website wants to increase its sales. Following is a sample of a typical interaction that takes place between a data scientist and the firm.

Data Scientist: What are your expectations from the data science project?

Company: We want to increase our daily online sales of our product “XYZ”.

Data Scientist: Which e-commerce platform are you using?

Company: We use three e-commerce platforms. Company also provides names of the three platforms.

Data scientist: How do you attract the customers for your product on e-commerce platforms?

Company: We promote (advertise) our goods on an e-commerce platform for relevant customers based on data from the platform's analytic engine.

Data scientist: What are the constraints?

Company: Advertising budget is limited.

Data scientist: What relevant data do you have available?

Company: We get sales history data which includes the consumer's profile as well as the date and time of purchase.

Data scientist: Do you have similar data of all the products sold by you through e-commerce platforms?

Company: Yes

Data Scientist: Do you obtain data of all the customers who access the e-commerce platform?

Company: We only get information about the number of users who access

platforms and when customers clicked on the e-commerce site, how long they were on the platform, what time of day resulted in the most successful transaction for any product, and so on.

Data scientists and businesses often communicate in this way, but this is only a sample. More in-depth communication is often necessary. Furthermore, data scientists benefit greatly from seeing the relevant business process in order to have a better grasp of the problem at hand, as well as any physical limitations, data collection and storage methods, etc.

This contact aids the data scientist in comprehending relevant business activity and identifying the problem. This additionally aids in the formulation of the project's objectives. For example, the study aims to identify the time of day with the greatest sales potential and predict the effects of marketing expenditures and propose strategies accordingly. This also facilitates the development of a timeline and delineates resource necessities.

Check Your Progress 2

1. What is the purpose of the first step in the data science process, and what does it involve?

.....

.....

.....

2. Why is it important for data scientists to possess business acumen and communication skills in the early phase of a project?

.....

.....

.....

3. In the e-commerce example, what kind of data was available to the data scientist, and how did this data help define the project goal?

.....

.....

.....

4. What is a data science project charter, and how does it contribute to project success?

.....

.....

.....

5. Why is it important for a data scientist to observe the relevant business process directly?

.....

6. What role does a proof of concept play in the goal identification phase of a data science project?

1.3.2 Data Collection

The next stage in your data science project is to gather the relevant data. Data scientists typically encounter two types of situations:

- Adequate data exists inside the organisation.
- No data is available within the organisation, or the existing data is insufficient.

In the first scenario, data is often accessible in the organization's formal repositories, including databases, data marts, data warehouses, and data lakes. These repositories are often managed by a team of IT specialists within the organisation. Some of the necessary data may also reside as Excel files, pdf documents, photos, etc., on the desktop of a domain expert.

Databases	It retains the data essential for application operation, primarily in its raw and unprocessed state.
Data warehouses	It aggregates structured data from several sources into a centralized repository, facilitating advanced analytics and reporting.
Data lakes	It is a massive data repository that often retains significant volumes of data in its original format, encompassing unstructured data such as chat logs and emails.
Data marts	It retains processed and validated data often utilised for specific reporting and analytical purposes. It may rely on warehouse data or function independently.

Accessing data from above repositories might be restricted by certain policies of the organization and may consume significant time in accessing. Data scientists must factor the same in its project timeline in step 1.

In circumstances when data is either unavailable or only available partially within the organisation, it may be necessary to gather information from the field or propose a template for the data collecting process, suitable sensors, and other relevant tools for the company. Many organisations are providing high-quality data for public and commercial use at no cost. Data scientists may also utilise such data provided by external parties.

Example: In case of the problem discussed in step 1, various data will be required. A sample of the data file is shown below.

Customer ID	Age	Gender	Access start date	Access start Time	Access end time	Product ID ordered, if any
Customer 1	43	M	12/12/2024	03:32:25	03:40:40	Nil
Customer 2	35	M	12/12/2024	12:12:40	12:40:20	12438
...	...	...	...	...	...	...
Customer 1	28	F	25/12/2024	09:32:12	10:10:10	23487

Similarly, other data like advertisements used by the company will also be collected by the data scientist.

☞ Check Your Progress 3

1. What are the two typical data availability scenarios faced by data scientists within an organisation?

.....

.....

.....

2. What are some common formal data repositories used within organisations, and what types of data do they store?

.....

.....

.....

3. What are some challenges data scientists may face when accessing internal data repositories?

.....

.....

.....

4. What steps can be taken if organisational data is unavailable or insufficient?

.....

.....

.....

5. Give an example of data that might be collected in a data science project, as mentioned in the text.

.....

.....

.....

1.3.3 Data Preparation

This is the step where we prepare the retrieved data for exploratory analysis and modelling. It involves cleaning the data by removing errors in your data, combining data retrieved from different sources, and converting the data into a certain shape for all further analysis. Data cleaning ensures that errors, inconsistencies, and outliers are removed. Data integration combines datasets from different sources, while data transformation prepares the data for modelling by reshaping variables or creating new features.

Data cleaning

Let us take few examples of errors in data:

Data entry error: Let us see following table indicating ten of students in a class:

Serial No.	1	2	3	4	5	6	7	8	9	10
Height (in cm)	160	156	168	145	15	165	160	155	149	147

If you have larger datasets you might want to run diagnostics tests as manual checks are tedious or not feasible in most cases. For example, a scatter plot and distribution plots may point out outliers. These outliers can be then checked manually. Remember, outliers need not be an erroneous entry always. Sanity checks are also performed to check if the values are physically or theoretically feasible. For example, people height more than 8 ft or less than a certain value for a certain age group might alert data entry mistakes.

Other errors include capital letter mismatch, impossible values, missing values, units of measurement mismatch, etc. Missing values in data are normally handled by data imputation. Data imputation is a statistical technique that replaces missing values in a dataset with estimated values. Missing values can be estimated using various methods such as maximum or minimum value, mean or moving average value, median value and most frequent value.

Data combining

Data originates from multiple distinct sources. It varies in dimensions, classification, and composition, encompassing databases, Excel files, and text documents. Let us take an example of data obtained from two different sources for a data science project.

Machine	Month	Number of failures
M1	January	2
M2	February	2
M3	March	6

Machine	Month	Load
M1	January	30%
M2	February	40%
M3	March	80%

We may combine these two tables into one as follows:

Machine	Month	Number of failures	Load
M1	January	2	30%
M2	February	2	40%
M3	March	6	80%

Transforming data

The objective of data transformation is to turn raw data into a format conducive to analysis and modelling. The following outlines some of the frequently employed data transformation techniques.

Data normalization: Data normalization aims to convert all data variables into a given range. Some of the widely used techniques for normalization are discussed below.

Min-Max Normalization: This transforms the original data linearly.

Suppose that x_{min} is the minima and x_{max} is the maxima of an attribute

x is the value you want to normalize

x' is the new value you get after normalizing the old value.

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$$

Example: If values of a feature vary between 200-800 and its natural value is $x=300$, then the normalized value will be:

$$x' = \frac{300 - 200}{800 - 200} = 0.17$$

Z-Score Normalization: In this normalization method mean and standard deviation of a variable is used for normalization.

$$x' = \frac{x - \text{Mean of } x}{\text{standard deviation of } x}$$

Example: Normalize the following dataset using Z value normalization.

X	4	4	1	8	15	10	25	13	26	16
---	---	---	---	---	----	----	----	----	----	----

Mean of X= 8.12

Standard Deviation of X= 12.2

Thus, normalized values of X will be as follows:

X	4	4	1	8	15	10	25	13	26	16
X'	-0.34	-0.34	-0.58	-0.01	0.56	0.15	1.38	0.4	1.47	0.65

Data reduction: Dimensionality reduction of data involves picking a subset of pertinent features or characteristics or amalgamating numerous features into a singular feature.

Data discretization: Transforming continuous data into discrete categories or intervals. For example, age can be discretized into discrete groups such as child, young and old. Discretization also helps in data reduction.

☞ Check Your Progress 4

1. Normalize the following dataset using Min-Max and z-score normalization methods.

X	400	1100	500	550	400	430	560	1000	800	600
---	-----	------	-----	-----	-----	-----	-----	------	-----	-----

.....

.....

.....

1.3.4 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is a critical initial step in the data science process that involves examining and visualizing datasets to uncover patterns, spot anomalies, test hypotheses, and check assumptions using summary statistics and graphical representations. It serves as a foundation for understanding the underlying structure of the data, identifying important variables, and guiding subsequent modeling efforts. EDA typically involves techniques such as distribution analysis, correlation assessment, and outlier detection, often using tools like histograms, box plots, scatter plots, and heatmaps. By providing insights into the quality and nature of the data, EDA helps data scientists make informed decisions about data preprocessing, feature selection, and modeling strategies. Its role is not only diagnostic but also strategic, as it enables a deeper

Once you have all the data you need and have put it in the right format, the next step is to understand it and pick the right modelling methods. Exploratory Data Analysis (EDA) is normally used for this purpose. Even though the goal isn't to clean up the data, you will probably still find oddities and outliers that you missed in the previous steps. This will mean you have to go back to steps 1-3 and fix it. To look at data, EDA mostly uses graphical tools like scatter plots, histograms, and box plots. For example, one may plot the number of units sold from January (month 1) to December (month 12) for the data captured. A simple scatter plot is shown in figure 3. This may reveal that sales of this product are higher during September to January. It may be worth classifying the data into two groups: September to January and February to August. Similarly, data scientists may plot trends in sales with

time of the day. These will help to understand if particular time of the day has higher potential for converting the platform visit into sales.

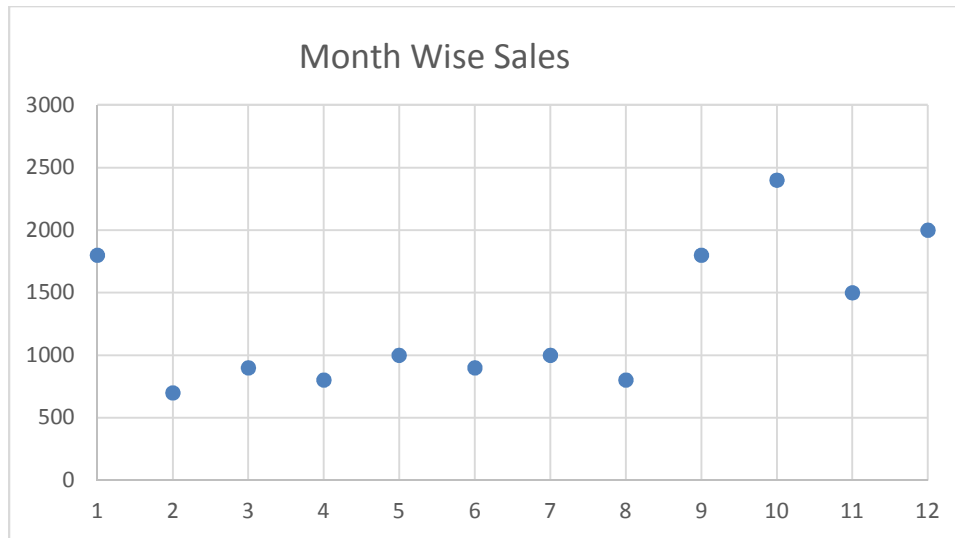


Figure 1.3: Scatter plot

In section 1.6 of this unit, the theory of EDA is explained in more depth.

1.3.5 Build Models

Predictions, object classification, or system comprehension, as sought in the first stage of data science, are the main aim of this stage. While the exploratory data analysis step focuses on identifying general trends in the data, this step is more focused on the goal of the overall data science project. In this step, machine learning or deep learning models are normally built to make predictions or classifications based on the data. This step heavily relies on statistics, data mining, and machine learning. The choice of algorithm depends on the complexity of the problem and the type of data. Important things to do in this stage include the following.

- Feature selection
- Model selection
- Model execution
- Model testing

Feature Selection: The procedure involves identifying and selecting the features or variables that most significantly contribute to the desired prediction variable or outcome. EDA outcomes can help you in this process. In examples discussed in 1.3.1 and 1.3.3, EDA may suggest that month of the year or time of the day as some of the important variables. Similarly, if the study aims to identify what advertisement is best during a particular time of the day, then one may study additional variables such as age and gender of customers. Also, interaction with domain experts will help you in identifying the right variables for your model. Presence of unnecessary variables or irrelevant variables can reduce model accuracy.

Model selection: Based on the purpose, two types of models—statistical and machine learning models—are used by data scientists. If the project aims to

find and explain the relationship between variables then statistical models are the best choice. For example, the city health administration department wants to identify high risk pregnancy in various Primary Health Centres (PHCs). It is crucial to comprehend the attributes of pregnant women to formulate intervention strategies that enhance care outcomes. In such cases data of pregnant women from PHC can be used to identify high risk women and identify appropriate treatment or preventive care interventions.

If the main purpose of the data science project is to make accurate predictions without focusing on pre-specified interactions, then ML (Machine Learning) models are normally used. For example, companies focus on personalized marketing by analysing a massive customer database. Such a database contains hundreds of variables and it's not easy to specify what variable defines a certain type of customer. ML models can segment customers into different types and predict segment membership of individuals with high accuracy, helping in personalized marketing.

Although some statistical models can make predictions, such models cannot capture complex relationships between data. Consequently, these statistical models typically exhibit low accuracy. Conversely, machine learning models can yield superior predictions; nevertheless, they are more challenging to comprehend and explain.

Once you select the y type of model, the next task is to select specific methods or algorithms for the problem. For example, one may choose supervised, unsupervised or reinforced learning ML models.

Supervised Learning Algorithm: This category of algorithms employs labelled datasets to train the algorithm for prediction and pattern recognition. Labelled data consists of examples with the correct answer or classification.

Consider a scenario in which a company seeks to forecast the Remaining Useful Life (RUL) of a cutting tool utilised in manufacturing equipment. The company is recording force and vibration until the cutting tool fails. The dataset established in this instance is a labelled dataset, as it includes the answer (RUL) for each cutting cycle of the tool, along with numerous variables pertaining to force and vibration from its previous operations. A supervised learning model can be constructed to forecast Remaining Useful Life (RUL) by inputting the current values of cutting cycle, force, and vibration for a new process. Some of the supervised learning algorithms are:

- Linear Regression
- Random Forest
- Support Vector Machines

Unsupervised Learning: Unsupervised learning uses unlabelled datasets to train algorithms to predict outcomes and recognize patterns. This means that there is no supervision. Interactions and connections between the presently available data points help in learning of the algorithm.

Some of the unsupervised learning algorithms are:

- K-means Clustering

- KNN (k-Nearest Neighbors)
- Hierarchical Clustering
- Anomaly Detection

Reinforcement Learning: In this type of algorithm system learns from its mistakes and improves over time. It uses dynamic datasets. The Following are some examples of reinforcement learning.

- Q-Learning
- Deep Q Network

This analogy helps to summarise the situation: *supervised learning occurs when children learn from their parents. Unsupervised learning occurs when children learn without adult supervision. Reinforced learning occurs when children learn from their errors.*

Model execution: Once the type of model has been decided, it is time to code it. You will have to test multiple models for accuracy and select the one that provides the greatest performance for the given situation. Coding can be a difficult process. Fortunately, most programming languages, like Python, already have libraries like StatsModels and Scikit-learn. These programs employ several of the most popular methods. These libraries can speed up model execution.

Model testing: Upon the development of the model, it will be necessary to evaluate its performance. Depending on the model type, some of the indices such as accuracy, false positives, false negatives, Mean Square Error (MSE), R, and modified R can be calculated for checking the performance of the model. If the desired performance is not attained, you may revert to the model and feature selection phase to select an alternative data model or add or remove features, and subsequently retest the model.

1.3.6 Model Deployment and Presentation

Once the analysis is completed and insights have been derived, the findings are communicated to stakeholders through reports, dashboards, or presentations. This stage plays a vital role in decision-making and often leads to the deployment of data science models into real-world environments. Deployment involves integrating these models with existing operational systems, such as enterprise software, web applications, or databases. However, this integration is not always straightforward and may present a range of technical and organizational challenges.

Some common challenges in model integration include:

- *Legacy software architectures:* Many organizations rely on outdated systems that were not designed with modern data science applications in mind. These legacy systems can pose compatibility issues, making it difficult to incorporate new models without significant refactoring or middleware development.
- *Lack of standardized data formats:* Data collected and stored across various departments or systems may follow inconsistent structures,

naming conventions, or encoding formats. This lack of standardization hinders seamless data flow and model interoperability.

- *Absence of standardized architectures:* Inconsistent or ad-hoc infrastructure setups can result in inefficiencies or duplicated efforts when attempting to integrate new components. Without a unified architecture, maintaining and scaling deployed models becomes more complex.
- *Resistance to change among stakeholders:* Organizational inertia and reluctance from employees or managers to adopt new technologies can delay or derail implementation efforts. This resistance may stem from a lack of understanding of the model's benefits, fear of job displacement, or concerns about reliability.

To overcome these challenges, several best practices can be adopted:

- *Standardization of data and architecture:* Establishing consistent data formats and adopting standardized architectural frameworks (such as microservices or containerized environments) can greatly ease integration and future scalability.
- *Development of Application Programming Interfaces (APIs):* APIs serve as bridges between models and operational systems, enabling secure and efficient data exchange. Well-documented APIs can also simplify maintenance and upgrades.
- *Collaboration between IT and data science teams:* Effective integration requires close cooperation between those who build the models and those who manage the infrastructure. Cross-functional teams can address issues proactively and ensure that the solution aligns with technical and business requirements.
- *Phase-wise implementation:* Deploying models in stages—starting with pilot projects or limited-scope applications—can help identify issues early, build confidence, and refine processes before full-scale rollout.
- *User training and change management:* Providing training to end-users and stakeholders ensures that they understand how to use and benefit from the new system. Change management initiatives, including workshops and feedback sessions, can ease transitions and improve acceptance.

In addition, the soft skills of data scientists, such as communication, empathy, and storytelling, play a crucial role. By clearly articulating the value of data-driven insights and addressing concerns proactively, data scientists can build trust and encourage buy-in from diverse stakeholders, facilitating smoother adoption and long-term success.

☞ Check Your Progress 5

- 1: What is the purpose of Exploratory Data Analysis (EDA) in the data science process, and what tools are commonly used for it?

.....
.....

2: How can EDA help identify trends in sales data, and what might be a practical application of such analysis?

.....

.....

.....

3: What is the main aim of the modeling stage in the data science process, and how does it differ from Exploratory Data Analysis (EDA)?

.....

.....

.....

4: What are the key activities involved in the modeling stage, and why is feature selection important?

.....

.....

.....

5: When should statistical models be preferred over machine learning models, and what is an example of such use?

.....

.....

.....

6: What is supervised learning, and how does it differ from unsupervised and reinforcement learning?

.....

.....

.....

7: What is the role of model testing, and what metrics are used to evaluate model performance?

.....

.....

.....

8: Why is the choice of algorithm important in the modeling stage, and what factors influence this choice?

.....

.....

.....

.....

9: How do programming libraries like Stats Models and Scikit-learn support the model execution process?

.....

.....

.....

10: What is the role of domain expertise in the feature selection process, and why is it important?

.....

.....

.....

11: Why is the communication of findings an important stage in the data science process?

Answer: Communicating findings is essential because it enables stakeholders to understand

.....

.....

.....

12: What are some common technical challenges faced during the integration of data science models into operational systems?

.....

.....

.....

13: How can organizational resistance affect the deployment of data science models, and what are some causes of this resistance?

.....

.....

.....

14: What best practices can help ensure successful integration and deployment of data science models?

.....

.....

.....

15: Why are soft skills important for data scientists during the deployment phase, and which skills are especially valuable?

.....

.....

1.4 DATA SCIENCE Vs DATA ANALYTICS

Data science and Data analytics are interconnected fields, both focuses on working with data and gaining insights. However, the focus of both is different. The insights drawn through Data science are more future looking while Data analytics focuses on gaining insights for the current period based on the past data. Data science is a broad field that includes Data analytics as well as mathematics, computers, statistics, programming, and machine learning.

We can say that both Data science and Data analytics are closely related disciplines that revolve around working with data to extract insights and support decision-making. However, while they share some tools and techniques, their objectives, scope, and approaches differ significantly.

In the broader sense we may describe or understand Data analytics and Data Science as follows:

Data analytics primarily focuses on examining historical and current data to identify trends, patterns, and insights that inform present-day decisions. It is often descriptive or diagnostic in nature, answering questions such as “What happened?” and “Why did it happen?” Organizations use data analytics to improve operational efficiency, optimize marketing campaigns, enhance customer experience, or monitor financial performance.

Data science, on the other hand, is a broader and more interdisciplinary field that not only includes data analytics but also integrates aspects of mathematics, statistics, computer science, and machine learning. It aims to uncover deeper insights, build predictive models, and develop algorithms that can forecast future events or automate complex processes. Data science tackles questions like “What will happen in the future?” and “What actions can we take to achieve a desired outcome?”

The following table summarizes the key differences between the two fields, with additional details and examples:

Features	Data analytics	Data science
Type of data	Deals with structured data only	Handles structured and unstructured data both
Type of intervention	Intervention is mostly done for problem solving through data analysis	Intervention is done not only to solve the problem but also to identify the problem in the first place.
Modelling and analysis	Mostly statistical analysis using statistical inferences	Predictive modelling using machine learning
Scope of analysis	Mostly focused on present time	More future looking
Ultimate Goal	To provide actionable insights that inform current decisions	To build intelligent systems and predictive models that

	and improve performance in the short term.	can learn and adapt over time , often supporting long-term strategy.
Example – Use Case	Analyzing monthly sales data to determine which products underperformed.	Creating a machine learning model that predicts which products customers are likely to buy based on browsing history and demographics.

☞ Check Your Progress 6

1: Differentiate between Data Science and Data Analytics

.....

.....

.....

1.5 TYPES OF DATA ANALYSIS

Data analysis can be categorized into four main types, each serving a distinct purpose in the decision-making process:

The four types of data analysis are:

- Descriptive Analysis
- Diagnostic Analysis
- Predictive Analysis
- Prescriptive Analysis

1. Descriptive Analysis

Descriptive analysis aims to summarize and interpret historical data to understand what has happened in the past. It is the simplest form of data analysis, using data aggregation and visualization techniques such as charts, graphs, and tables to present trends, averages, and patterns. This type of analysis helps organizations track performance over time and report on key metrics. For example, a retail company might analyze sales data from the last quarter and find that sales increased by 15% in December compared to November. Dashboards showing total sales, customer visits, and product performance are typical examples of descriptive analysis.

2. Diagnostic Analysis

Diagnostic analysis goes a step deeper by investigating the causes behind trends or outcomes identified during descriptive analysis. Its purpose is to answer the question, “Why did it happen?” This analysis involves techniques like correlation analysis, root cause analysis, and data drilling to identify relationships and anomalies in the data. For instance, if a digital marketing team notices a dip in website traffic in March, they might perform diagnostic analysis and discover that a major advertising

campaign ended in February, which led to reduced visibility and fewer visits.

3. Predictive Analysis

Predictive analysis focuses on forecasting future events or trends based on historical data. It employs statistical modeling, data mining, and machine learning algorithms to estimate the likelihood of future outcomes. This enables organizations to anticipate changes and make informed decisions. For example, a bank may use a predictive model to identify customers who are likely to default on loans by analyzing their credit history, income, and past behaviors. Such predictions allow the bank to manage risk and take preventive measures.

4. Prescriptive Analysis

Prescriptive analysis takes predictive insights further by recommending specific actions to achieve desired goals. It combines optimization, simulation, and decision modeling techniques with business rules and constraints to suggest the best course of action. For example, an airline might use prescriptive analysis to determine the best pricing strategy for flight tickets. By considering demand forecasts, competitor pricing, and booking patterns, the model recommends ticket prices that maximize revenue.

Together, these four types of data analysis form a hierarchy: descriptive analysis explains what happened, diagnostic analysis explains why it happened, predictive analysis estimates what might happen, and prescriptive analysis advises on what actions should be taken.

1.5.1 Descriptive Analysis

The first kind of data analysis is known as descriptive analysis. The "what happened" question can be answered via descriptive analysis, which summarises previous data.

Think about an engineering institute that wants to analyse how many female students are enrolled in three different engineering majors. The university looked at admittance statistics from three different engineering fields during the last five years. The data are shown below.

	2020		2021		2022		2023		2024	
	Total students	Female students	Total students	Female students	Total students	Female students	Total students	Female students	Total students	Female students
Computer Science and Engineering	50	6	50	6	50	6	100	40	100	40
Electrical Engineering	50	5	50	6	50	5	100	30	100	32
Mechanical Engineering	50	5	50	6	50	5	100	28	100	28

For the first three years, descriptive analysis shows that the percentage of female students was around 10-12% across all majors. Over the following two years, the percentage of female students rose across all majors and

reached around 28-40%. Compared to the other two fields, computer science and engineering saw the greatest increase.

Similarly, descriptive analysis can be employed in different industries to monitor Key Performance Indicators (KPIs) such as monthly revenue, number of new technologies developed, number of complaints resolved, and total products produced.

1.5.2 Diagnostic Analysis

"Why did it happen?" can be answered using diagnostic analysis. The results of the descriptive analysis form the basis of diagnostic analysis, which seeks to identify the underlying causes of the results. In the example given in the previous section, the school may be interested to know why the number of female students rose throughout the past two years. Additional data pertinent to the issue, including hostel facilities, scholarships for female students, transport services, and government policies, may be analysed to ascertain the underlying reasons.

Other examples of diagnostic analysis include investigating the reasons for quality rejections, the reasons for a reduction in throughput, the reasons for high employee attrition in a particular year, determining the reasons why certain months have higher or lower traffic on an e-commerce website, and determining the reasons for patient readmission after surgery in a week.

1.5.3 Predictive Analytics

"What is likely to happen?" can be answered with predictive analysis. It guesses what will happen in the future by using classification models, regression models, grouping models, time series models, or neural networks models.

In the previous example, schools might want to predict the number of female students in the next 2 years.

Similarly, the following are some examples of predictive analysis in industries.

- Predicting the remaining useful life of an equipment,
- Predicting customer churn (customer leaving),
- Forecasting demand for products,
- Predicting the likelihood of loan defaults to make informed lending decisions

1.5.4 Prescriptive Analysis

The question "What should we do next?" is addressed through prescriptive analysis. It suggests the best course of action based on predictions. In the previous example, prescriptive analysis can suggest actions such as best outreach activity to attract female students for various courses.

Similarly, in the case of example presented in section 1.2, prescriptive analysis can help the doctors in deciding the best weaning off strategy. Other

applications of prescriptive analysis can be in the field of personalised education where based on the past academic grades, learning pattern and other data of students, schools can develop a personalised teaching methodology for different groups of students.

While prescriptive analysis is normally the end goal of any data science project other analyses are important to reach to this level.

☞ Check Your Progress 7

1. Describe various data analysis techniques.

.....

.....

.....

.....

1.6 PHILOSOPHY OF EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) is a foundational and critical phase in the data analysis process. It involves examining the dataset in detail before making any assumptions or applying formal statistical models. The core objective of EDA is to understand the structure, distribution, and relationships within the data. By doing so, it helps detect anomalies such as missing values or outliers, identify patterns and trends, uncover relationships between variables, and assess data quality. These insights are essential for guiding further analysis, including the selection of appropriate models and statistical tools.

*EDA methods can be broadly classified into graphical and non-graphical approaches, and they can be applied in either univariate or multivariate contexts. **Univariate analysis**, the simplest form of EDA, deals with a single variable at a time. Since it involves only one variable, it does not provide insights into causal relationships or associations. The main goal of univariate analysis is to describe the characteristics of the variable and identify any underlying patterns. **Non-graphical univariate methods** include numerical summaries such as measures of central tendency (mean, median, mode), dispersion (range, variance, standard deviation), and distribution shape (skewness, kurtosis). However, relying solely on numerical summaries may not provide a complete picture. Therefore, graphical methods such as stem-and-leaf plots, histograms, and box plots are often employed to visually represent data distributions, spot outliers, and gain intuitive understanding of the variable.*

When multiple variables are involved, **multivariate analysis** is used to explore relationships among them. This type of analysis helps in understanding how variables interact with each other and whether any meaningful associations exist. *Non-graphical techniques for multivariate analysis* include cross-tabulation, which is useful for examining the relationship between categorical variables, and statistical measures such as

correlation coefficients and covariance, which quantify the strength and direction of relationships between numerical variables. Graphical methods are also widely used in multivariate EDA. Grouped bar charts enable comparisons across different categories and sub-categories, while scatter plots visually depict the relationship between two continuous variables, highlighting patterns, trends, and potential clusters. Heat maps are another powerful tool, representing data values with color gradients to quickly show correlations or intensities across multiple variables.

Overall, EDA plays an essential role in converting raw data into meaningful insights. By combining graphical and non-graphical methods and exploring both individual variables and their interactions, analysts are better equipped to make informed decisions, refine hypotheses, and choose appropriate analytical models for subsequent stages of data analysis.

In short, it can be understood that Exploratory data analysis, or "EDA," constitutes a crucial initial phase in data analysis. The primary objective of EDA is to examine data prior to forming any assumptions. It can assist in identifying errors, comprehending patterns within the data, detecting outliers and uncovering intriguing relationships among the variables. It also aids in the initial selection of suitable models and in identifying correlations between the explanatory and outcome variables. Exploratory Data Analysis (EDA) can be conducted using graphical or non-graphical methods. Additionally, it can be classified as either univariate or multivariate. Accordingly various EDA techniques are discussed below.

Univariate analysis:

This represents the most basic form of data analysis, wherein the dataset comprises a single variable. As it pertains to a single variable, it does not address causation. The primary objective of univariate analysis is to characterise the data and identify inherent patterns. This can be graphical or non-graphical. Non-graphical analysis includes measurements of central tendency (such as the mean or median), measures of dispersion (such as the range or standard deviation), and measures of distribution form (such as skewness or kurtosis). It is normally difficult to get a full picture of the data from non-graphical methods. Graphical methods such as stem-and-leaf plots, histograms, Box plots can be used to get a better picture.

Multivariate analysis:

Multivariate analysis is employed when data encompasses multiple variables. It often illustrates the correlation among two or more variables. The relationship can be illustrated using non-graphical methods such as cross-tabulation and statistics (correlation and covariance for example), or through graphical methods such as grouped bar plots or bar charts, where each group signifies one level of a variable and each bar inside a group denotes the levels of the other variable. Likewise, a scatter plot is employed to graph data points on a horizontal and vertical axis to illustrate the extent to which one variable influences another. A heat map is another type of multivariate graphical exploratory data analysis, representing data visually with values indicated by colour.

1.7 ROLE OF DATA SCIENTIST AND DATA ANALYST

Key differences between Data science and Data analytics discussed in section 1.4. It dictates the differences in the role of a Data scientist and Data analyst.

Companies use Data analysts to address their business problems. Typical activity performed by Data analysts is finding trends in sales or usage. Data scientists are hired to work closely with domain experts not only to solve the problem but also to identify the problem. As Data scientists are required to work closely with domain experts as well as top management, they are required to have strong communication and interpersonal skills. Data analysts jobs are mostly confined to analyzing data. They will be less involved in data driven decision making within the organization. Data scientists are normally a part of the decision-making team in the organization. Therefore, storytelling and management skills are required in data scientists to translate findings into business strategies.

On the technical side, Data analysts mainly require knowledge of algebra, probability and statistics. On the other hand, data scientists furthermore demand expertise in machine learning to develop robust predictive models. As data analytics mainly handles structured data, they make use of SQL queries to retrieve and manage data. Data scientists also use NoSQL for unstructured data. Relatively stronger programming skills are expected from data scientists.

Considering the above it is normally observed that Data scientists hold advanced degrees like Master or PhD in data science or relate subject. On the other hand, one can gain required skills for data analysts as a part of their regular undergraduate skill or through some additional courses.

The role, knowledge, skill required from a Data scientist is significantly higher than a Data analyst. The same also reflects in average salary for these two roles across various sections.

1.8 CROWDSOURCING

Data science utilises data to address business challenges. Frequently, adequate data are either unavailable or insufficient to derive the necessary conclusions. Organisations implement crowdsourcing strategies to gather information or opinions from a vast population that contributes data through the internet, social media, and mobile applications. Crowdsourcing enables companies to conserve time and resources by tapping into people from across the world. Data crowdsourcing can enhance customer service, elucidate client requirements, and facilitate superior company decisions.

For example, Google Maps, while used to navigate on the road, sometimes asks users inputs about construction work going on or large objects on the road. Similarly, many organizations collect feedback through online surveys from customers to improve their product or services.

Crowdsourcing can be very effective as it uses the power of a large group of people in gathering data. Nonetheless, the quality of the data remains a concern, as crowdsourced information is often subject to biases. Quality control techniques, including proofreading and validation, are crucial for ensuring the accuracy of data obtained through crowdsourcing.

So, we can understand that Crowdsourcing in data science plays a crucial role, especially when data is insufficient or unavailable to address business challenges effectively. In such cases, organizations turn to crowdsourcing as a strategic solution for gathering data and insights from a broad audience. This approach leverages the collective knowledge and participation of people across the globe who contribute through internet platforms, social media, and mobile applications. By tapping into the power of the crowd, organizations can collect vast amounts of data quickly and cost-effectively. This process not only saves time and resources but also extends the reach of data collection across diverse geographical and demographic groups, making the data more representative.

Practical applications of crowdsourcing are evident in everyday tools and services. For example, Google Maps relies on users to report real-time issues such as roadblocks or construction, allowing the system to adjust navigation routes more accurately. Similarly, companies often use online surveys and feedback forms to gather customer input, which helps them refine their products and improve customer service. These methods allow businesses to gain deeper insights into client needs and make informed, data-driven decisions.

The advantages of crowdsourcing in data science are significant. It enables scalability by allowing organizations to gather large-scale data efficiently. It also brings in diversity, offering varied perspectives and data points that might not be accessible through traditional means. Furthermore, crowdsourcing often sparks innovation, as user-generated content and feedback can uncover hidden issues or inspire new features and ideas that internal teams might overlook.

However, crowdsourcing is not without its challenges. One of the most pressing concerns is data quality. Since contributors come from different backgrounds and skill levels, the data they provide can be biased, inaccurate, or inconsistent. Some users may offer incomplete or even intentionally misleading information. As a result, organizations must implement robust quality control measures to ensure the reliability of the collected data. Techniques such as proofreading, cross-validation, algorithmic filtering, and contributor reputation systems are essential to minimize errors and ensure data accuracy.

Given its strengths and limitations, crowdsourcing should be viewed as a complementary tool in data science rather than a standalone solution. It is most effective when tasks require subjective judgment, diverse perspectives, or quick, large-scale data collection that automated systems cannot achieve alone. Ethical considerations are also crucial—organizations must ensure transparency, protect user privacy, and offer value or incentives to contributors for their participation.

Overall, crowdsourcing is a powerful method of data collection that, when used responsibly, can significantly enhance the scope and quality of data science initiatives. Its success hinges on a careful balance between speed, scale, and data quality, supported by well-designed processes, validation mechanisms, and ethical practices. When implemented thoughtfully, crowdsourcing enables organizations to derive meaningful insights, make better decisions, and innovate with greater agility.

1.9 LANGUAGES FOR DATA SCIENCE

Data science is a multidisciplinary field that involves extracting meaningful insights from structured and unstructured data. To carry out various tasks such as data cleaning, exploration, visualization, modelling, and deployment, data scientists rely heavily on programming languages. The choice of language often depends on the specific task, the volume of data, system performance requirements, and community support. Several programming languages have emerged as essential tools in the data science ecosystem, each with its own strengths and use cases.

Over the last one decade many programming languages, libraries and software tools have been developed to help data scientists in implementing data science projects. Though not exhaustive, the following list provides a glimpse of some of such widely used programming languages, tools and libraries. They provide many utility features which help us to complete most of the complex tasks efficiently within a very limited time.

1. **Python:** Python is the most popular and widely used programming language in data science. Its simplicity, readability, and a vast ecosystem of libraries make it ideal for both beginners and professionals. Libraries such as **NumPy** and **Pandas** are used for numerical computations and data manipulation, **Matplotlib** and **Seaborn** for data visualization, and **Scikit-learn**, **TensorFlow**, and **PyTorch** for machine learning and deep learning. Python's versatility also allows integration with web applications, APIs, and big data tools, making it a comprehensive choice for end-to-end data science projects.
2. **R:** R is another powerful language tailored specifically for statistical computing and data analysis. It is preferred by statisticians and academic researchers due to its strong capabilities in handling data visualization and statistical modeling. Packages like **ggplot2**, **dplyr**, and **caret** provide powerful tools for data wrangling, visualization, and modeling. R has an active community and extensive documentation, making it a go-to language for advanced statistical analysis and academic research in data science.
3. **SQL (Structured Query Language) :** SQL is indispensable for data extraction and manipulation, particularly when working with relational databases. Data scientists often need to retrieve, filter, aggregate, and join data stored in databases, and SQL provides an efficient way to perform these tasks. Even though it is not used for statistical modeling or machine learning, SQL is often the first step in the data pipeline, making it a foundational language for any data science workflow.

4. **Java and Scala:** Java and Scala are widely used in big data environments. Tools like **Apache Hadoop** and **Apache Spark**, which are critical for handling large-scale data processing, are often implemented using these languages. While Java offers robustness and wide platform support, Scala, being more concise and functional, is preferred for Spark-based data science applications. These languages are particularly useful in enterprise environments where performance and scalability are crucial.
5. **Julia:** Julia is a high-performance programming language that is gaining popularity in the data science and scientific computing communities. It is designed for numerical and computational work and offers speed comparable to lower-level languages like C, while maintaining the ease of use of higher-level languages like Python. Julia is particularly suited for large-scale simulations, optimization problems, and complex mathematical modeling.
6. **SAS (Statistical Analysis System):** SAS is a proprietary language used primarily in industries like healthcare, finance, and banking, where security and regulatory compliance are important. It offers robust tools for statistical analysis and data management but is often seen as less flexible compared to open-source alternatives like R and Python.
7. **MATLAB:** MATLAB is used for numerical computing, simulations, and algorithm development, especially in engineering and scientific research. While not as commonly used in mainstream data science, it is still relevant in specialized fields where matrix operations and simulations are required.

Choosing the right programming language in data science depends on the problem at hand, the domain of application, and the tools required. **Python** remains the dominant language due to its broad applicability and community support. **R** excels in statistical tasks, **SQL** is essential for database interactions, and languages like **Java**, **Scala**, **Julia**, and **MATLAB** serve specific high-performance or domain-specific needs. A proficient data scientist often has working knowledge of multiple languages and selects the most appropriate tool for each stage of the data science process.

1.10 APPLICATIONS OF DATA SCIENCE

Data science is widely applied across various sectors, helping professionals make better decisions using insights drawn from data. Understanding these applications gives us a clearer picture of how data science adds value in real-world scenarios.

Education Sector: In the education sector, data science is used to improve both teaching and learning. Teachers can analyze student feedback to understand what works and what doesn't, allowing them to adjust their teaching strategies accordingly. Data about students' academic history and learning patterns can also be used to personalize the learning experience, catering to individual needs. School management teams can use data to

identify trends in admissions and develop strategies to attract more students, ensuring better resource planning and outreach.

Healthcare Sector: Data science plays a vital role in healthcare by supporting clinical decision-making. Doctors can use data-driven insights to diagnose and treat patients more effectively. Personalized and preventive care becomes possible by analyzing patient data, which helps in designing treatment plans tailored to each individual's health history. Insurance companies apply data science to develop personalized insurance policies based on customer profiles and risk factors. Additionally, hospitals, city administrators, and governments use data to track and predict disease outbreaks, enabling them to take preventive measures and allocate resources efficiently.

Manufacturing Industry: In the manufacturing industry, data science helps organizations improve productivity and reduce costs. Analyzing data from production processes allows companies to identify inefficiencies and schedule predictive maintenance to prevent equipment failure. Quality control also benefits from data science, as companies can analyze product defect data to find patterns and root causes, helping to reduce future quality issues. Supply chain management is another area where data science proves valuable—it helps identify bottlenecks, manage inventory more effectively, and streamline delivery processes to improve overall efficiency.

E-Commerce: In the world of e-commerce, data science is used to enhance customer experience by making personalized product recommendations. This is done by analyzing users' purchase history and search behavior, allowing platforms to suggest relevant products that increase user satisfaction and sales conversions.

Finance: In finance, data science is applied in various ways. In the stock market, for example, historical data is analyzed to predict future price trends, helping investors make more informed decisions. Banks and financial institutions also rely on data science to detect and prevent fraud. This includes identifying suspicious transactions or activities like rogue trading, speculative behavior, or violations of regulations, thereby ensuring greater security and trust.

Transport Industry: In the transport industry, data science contributes significantly to innovation and efficiency. One major application is in the development of autonomous or driverless vehicles. These vehicles rely on data to understand road conditions, predict the movement of other vehicles, and make real-time decisions. In aviation, data science helps airlines predict flight delays by analyzing weather conditions, traffic data, and operational patterns, allowing them to improve scheduling and passenger communication.

By exploring these examples, it becomes clear that data science is not limited to one field but has broad applications that impact daily life and business operations. As you continue learning, try to observe how data-driven decisions are being made in various sectors around you. This awareness will help you better understand the role of data science and inspire new ideas for how it can be applied.

 Check Your Progress 8

1. What is the primary purpose of Exploratory Data Analysis (EDA), and how is it typically conducted?

.....

.....

.....

.....

2. What distinguishes univariate analysis from multivariate analysis in EDA?

.....

.....

.....

.....

3. Why are graphical methods considered important in EDA, particularly in univariate analysis?

.....

.....

.....

.....

4. What are some common graphical techniques used in multivariate exploratory data analysis, and what do they illustrate?

.....

.....

.....

.....

5. Compare the role of Data Scientist and Data Analyst in any project

.....

.....

.....

.....

6. What is Crowdsourcing? Give its advantages and disadvantages.

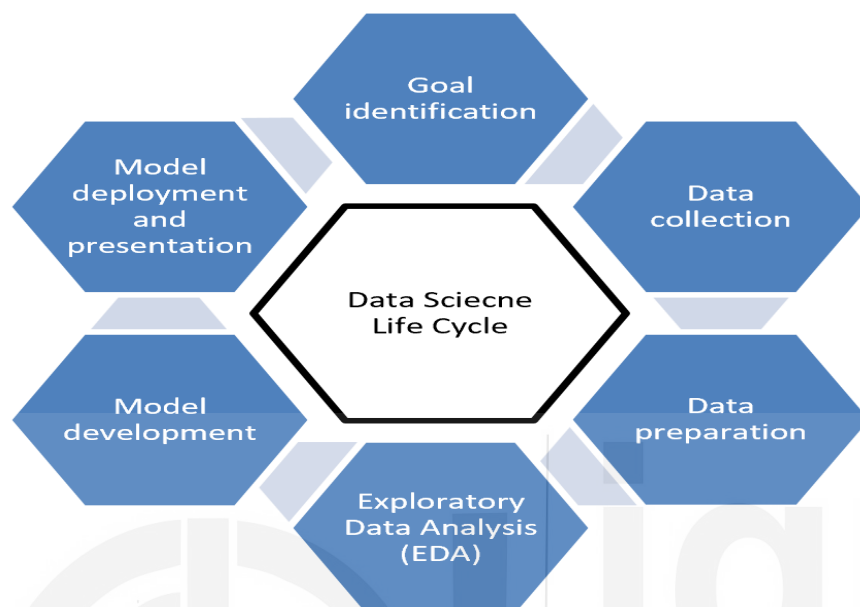
.....

.....

.....

1.11 DATA SCIENCE LIFE CYCLE

The Data Science Process follows multiple steps as discussed in section 1.3. These steps make the entire Life Cycle of any data science project.



A brief review of the steps involved in the data Science Life Cycle are as follows:

Step 1: Goal Identification- The first step involves clearly defining the business problem or objective. Understanding the goal helps in setting the direction for the entire project and ensures that the data science efforts align with the organization's needs. This stage often requires collaboration with stakeholders to clarify expectations and success criteria.

Step 2: Data Collection- In this phase, relevant data is gathered from various sources such as databases, APIs, web scraping, sensors, or external datasets. The quantity, quality, and diversity of collected data significantly influence the outcome of the project. Ensuring the data is reliable and relevant is key.

Step 3: Data Preparation- Data preparation, or data cleaning, involves processing the collected data to make it usable for analysis. This includes handling missing values, removing duplicates, correcting errors, and transforming data into suitable formats. Proper preparation improves the accuracy and efficiency of subsequent steps.

Step 4: Exploratory Data Analysis (EDA)- EDA is the process of summarizing and visualizing the data to uncover patterns, trends, and anomalies. It helps in understanding the underlying structure of the data, identifying relationships between variables, and detecting outliers, guiding feature selection and model choice.

Step 5: Model Development- During model development, various algorithms are applied to the prepared data to build predictive or descriptive

models. This includes selecting appropriate techniques, training models, tuning parameters, and validating performance using metrics to ensure the model effectively addresses the problem.

Step 6: Model Deployment and Presentation- The final stage involves integrating the developed model into a production environment where it can provide real-time or batch predictions. Additionally, findings are communicated to stakeholders through reports, dashboards, or presentations, often requiring clear visualization and storytelling to support decision-making.

Data scientists must follow all the steps to avoid mismatch of stakeholder expectation and the project outcomes. However, often it is required to go back to some of the previous steps to get a better understanding of data.

Check Your Progress 8

1. APAAR, which stands for Automated Permanent Academic Account Registry (APAAR), is a specialized identification system designed for all students in India. APAAR ID makes it possible to personalize teaching approaches. Inspired and facilitated by APAAR ID, school management is interested in adopting a personalized teaching approach to fit each student's needs by looking at their grades, learning patterns, etc. Discuss various steps of the data science life cycle for this personalized teaching project in detail.

.....
.....
.....

1.12 SUMMARY

This unit introduces the foundational concepts of Data Science, explaining its importance, processes, and applications. It distinguishes data science from related fields like data analytics and clarifies the roles of data scientists and analysts.

Key Points Covered:

- Introduction to Data Science: Data science is the systematic process of extracting actionable insights from large, diverse datasets (both structured and unstructured) using tools from mathematics, statistics, machine learning, programming, and domain expertise.
- Data Science Process & Life Cycle: The data science workflow is iterative and involves:
 1. Goal identification – understanding the business problem and project objectives.
 2. Data collection – gathering data from internal repositories or external sources.
 3. Data preparation – cleaning, integrating, and transforming data to make it suitable for analysis.

4. Exploratory Data Analysis (EDA) – using graphical and statistical tools to understand patterns, detect outliers, and guide model selection.
 5. Model development – building predictive or descriptive models using machine learning or statistical methods.
 6. Model deployment and presentation – integrating models into systems and communicating results to stakeholders.
- Data Science vs. Data Analytics: Data analytics focuses on interpreting past and current data (mostly structured) to solve specific problems, while data science is broader, predictive, and includes unstructured data, advanced modeling, and problem identification.
 - Types of Data Analysis:
 - Descriptive Analysis ("What happened?")
 - Diagnostic Analysis ("Why did it happen?")
 - Predictive Analysis ("What will happen?")
 - Prescriptive Analysis ("What should we do next?")
 - Philosophy of Exploratory Data Analysis (EDA): EDA involves understanding data's underlying structure using univariate and multivariate graphical/non-graphical methods to inform modeling decisions.
 - Roles of Data Scientist vs. Data Analyst: Data scientists possess deeper technical expertise and business acumen, working closely with domain experts to identify and solve problems, often holding advanced degrees. Data analysts focus on analyzing structured data and producing reports to support decisions.
 - Crowdsourcing: Crowdsourcing is used to gather large volumes of data or opinions from a wide audience to supplement organizational data, with emphasis on quality control.
 - Languages and Tools for Data Science: Common tools include Python (with libraries like pandas, NumPy, Scikit-learn, TensorFlow), R, SQL, MATLAB, Power BI, SAS, and platforms like Anaconda.
 - Applications of Data Science: Data science is applied in various sectors such as education (personalized learning), healthcare (decision support, personalized care), manufacturing (process optimization), e-commerce (personalized recommendations), finance (fraud detection), and transportation (driverless cars, flight delay prediction).

The unit ends with exercises, answers, and further reading recommendations to reinforce learning.

1.13 SOLUTIONS/ANSWERS

Check Your Progress 1

Question 1: What are key functional blocks of data science.

Answer: The key functional blocks of data science, as illustrated in the text, include:

- **Problem Definition** : Understanding the business or clinical need (e.g., optimizing weaning from mechanical ventilation) and Collaborating with domain experts to define goals and challenges.
- **Data Collection and Integration** : Includes Gathering data from multiple sources such as ventilators, monitors, lab tests, EHRs, etc, and Handling various data types: structured, unstructured, metadata, manual, and sensor-based.
- **Data Storage and Retrieval** : Involves Using tools like SQL, Python (pandas, NumPy), R to store and manage data.
- **Data Preprocessing and Visualization** : Relates to Cleaning, transforming, and visualizing data to detect trends, patterns, and anomalies.
- **Model Building** : Involves Applying techniques like regression, decision trees, clustering, and neural networks, and Using libraries like Scikit-learn, TensorFlow, and PyTorch.
- **Model Evaluation** : That is Assessing performance using metrics like accuracy, precision, and recall.
- **Insight Generation** : That is Producing actionable insights (e.g., reducing risk of VILI or improving detection).
- **Deployment** : Relates to Integrating models into real-world systems using tools such as Flask, Django, AWS, and Docker.

Question 2: What types of data are involved in critical care datasets and how are they categorized?

Answer: Critical care datasets involve **structured**, **unstructured**, and **metadata**.

- **Structured data** includes measurable values like temperature, blood pressure, and oxygen saturation.
- **Unstructured data** includes X-ray and CT scan images, reports, and handwritten clinical notes.
- **Metadata** includes details such as height and registration number. Data can be collected **manually** or via **digital sensors**, and may be recorded **continuously** or at **specific intervals** (e.g., weekly or hourly).

Question 3: How can data science support optimal mechanical ventilation and weaning in critical care?

Answer: Data science can assist experts by improving **detection of patient conditions**, enabling **timely intervention** and **personalized treatment**. This helps in reducing **Ventilator-Induced Lung Injury (VILI)** risk, improving **patient outcomes**, and optimizing **resource utilization**. Models built using data science can predict health conditions and suggest improved **weaning strategies**.

Question 4: What are the main tools and techniques used by data scientists for analyzing healthcare data in the given case?

Answer: Key tools and techniques include:

- **Languages & Libraries:** SQL, Python (pandas, NumPy), R
- **Modeling Techniques:** Regression, decision trees, clustering, artificial neural networks
- **ML Tools:** Scikit-learn, TensorFlow, PyTorch
- **Visualization:** Used to identify trends and anomalies
- **Deployment Tools:** Flask, Django, AWS, Docker- These help in storing, manipulating, analyzing, and deploying models for real-world healthcare applications.

Question 5: Why is collaboration with domain experts important in a data science project related to critical care?

Answer: Domain experts (e.g., doctors and healthcare professionals) help in:

- Defining problems and goals
- Understanding the data sources
- Interpreting the medical and physiological context (e.g., mechanical ventilation process)
- Addressing challenges like data privacy, sensor installation, and result accuracy their input ensures the relevance, accuracy, and applicability of the data science solutions in clinical settings.

👉 Check Your Progress 2

Question 1: What is the purpose of the first step in the data science process, and what does it involve?

Answer: The purpose of the first step, *Goal identification*, is to understand the business problem and define the objectives of the data science project. It involves clarifying the 'what' and 'why' of the project through communication with stakeholders. This includes understanding management's expectations, the broader context, project impact, intended use of outcomes, and existing constraints. A project charter is created to align all stakeholders and prevent project failure due to unclear objectives.

Question 2: Why is it important for data scientists to possess business acumen and communication skills in the early phase of a project?

Answer: Business acumen and communication skills are critical in the early phase because data scientists need to engage effectively with stakeholders to understand the business problem, ask the right questions, and identify constraints. These skills help ensure clear alignment on project goals, feasibility, and resource planning, which are necessary to set the project on the right path from the beginning.

Question 3: In the e-commerce example, what kind of data was available to the data scientist, and how did this data help define the project goal?

Answer: The data available included sales history (with customer profiles, date, and time of purchase), platform usage data (number of users, click activity, time spent), and performance data (time of day with most transactions). This data helped the data scientist understand consumer behavior, platform engagement, and marketing impact, enabling the formulation of goals such as identifying peak sales times and optimizing advertising strategies within a limited budget.

Question 4: What is a data science project charter, and how does it contribute to project success?

Answer: A data science project charter is a document that outlines the project's objectives, scope, constraints, resources, and alignment with business goals. It ensures that all stakeholders have a shared understanding of the project. This alignment is crucial for avoiding miscommunication, setting realistic expectations, and planning for necessary data and human resources, thereby increasing the likelihood of project success.

Question 5:

Why is it important for a data scientist to observe the relevant business process directly?

Answer: Observing the relevant business process helps the data scientist gain a deeper understanding of the actual workflow, physical limitations, data collection and storage methods, and other contextual factors. This direct insight enhances problem comprehension, ensures more accurate analysis, and leads to better-informed project objectives and solutions.

Question 6: What role does a proof of concept play in the goal identification phase of a data science project?

Answer: A proof of concept is used to demonstrate the feasibility of the proposed project during the goal identification phase. It helps validate whether the project's approach is likely to succeed and can guide decisions on continuing with full-scale implementation. It also provides early feedback, helping refine project scope, data requirements, and resource planning.

☞ **Check Your Progress 3**

Question 1. What are the two typical data availability scenarios faced by data scientists within an organisation?

Answer : Data scientists typically encounter two distinct scenarios when beginning the data collection phase of a project. In the first scenario, adequate data already exists within the organisation. This means that relevant and usable data is available in various forms and stored across different systems. In the second scenario, either no data is available at all, or the data that exists is not sufficient to address the specific requirements of the project. This distinction is crucial because it determines the data acquisition strategy that the data scientist must adopt.

Question 2. What are some common formal data repositories used within organisations, and what types of data do they store?

Answer : Within organisations, data is usually stored in structured repositories managed by IT specialists. These include databases, data warehouses, data lakes, and data marts, each serving different purposes. Databases primarily contain raw, operational data necessary for running applications, often in its unprocessed form. Data warehouses are designed to store structured data from multiple sources in a centralized format, supporting complex queries, reporting, and analytics. Data lakes, on the other hand, are large-scale repositories capable of storing vast volumes of data in its original format, including unstructured data types such as email logs, chat records, and sensor outputs. Lastly, data marts are subsets of data warehouses or independent repositories that contain cleaned and validated data tailored for specific reporting or analytical tasks.

Question 3. What are some challenges data scientists may face when accessing internal data repositories?

Answer: Accessing data within an organisation is not always straightforward. One of the main challenges is that data repositories are often governed by internal policies that restrict who can access what data and under what conditions. As a result, obtaining access may require formal approvals or the involvement of IT departments, which can be time-consuming. Additionally, if some data resides outside central repositories—such as in Excel files, PDFs, or images stored on individual desktops—it may be difficult to locate or standardise. Data scientists must account for these potential delays and hurdles during the initial planning stage of their project to avoid disruptions later on.

Question 4. What steps can be taken if organisational data is unavailable or insufficient?

Answer : In cases where data is either not available or not sufficient within the organisation, data scientists must explore alternative ways to gather the required information. This could involve going out into the field to collect primary data, designing data collection templates, or recommending appropriate sensors and tools to capture relevant data. Additionally, data scientists may leverage publicly available datasets provided by external sources such as government agencies, research institutions, or open-data platforms. Many of these external datasets are of high quality and can be used freely for both academic and commercial purposes, making them a valuable resource when internal data is lacking.

Question 5. Give an example of data that might be collected in a data science project, as mentioned in the text.

Answer : An example of data that could be collected during a data science project includes customer interaction logs with specific details such as Customer ID, Age, Gender, Access Start Date and Time, Access End Time, and the Product ID of any items ordered. This kind of data helps in understanding user behavior and preferences. For instance, the log could

show how long a customer stayed on a platform and whether they made a purchase during that session. In addition to access logs, the data scientist may also collect other contextual information, such as which advertisements the company used during the period, to explore correlations between marketing campaigns and customer behavior. Such comprehensive data collection supports more accurate analysis and model development.

☞ Check Your Progress 4

Question 1 : Normalize the following dataset using Min-Max and z-score normalization method.

X	400	1100	500	550	400	430	560	1000	800	600
---	-----	------	-----	-----	-----	-----	-----	------	-----	-----

Solution: Normalization of Dataset and Interpretation

$X = [400, 1100, 500, 550, 400, 430, 560, 1000, 800, 600]$

Min-Max Normalization : Formula: $X_{\text{norm}} = (X - X_{\text{min}}) / (X_{\text{max}} - X_{\text{min}})$

Where: $X_{\text{min}} = 400$, $X_{\text{max}} = 1100$, $(X_{\text{max}} - X_{\text{min}}) = 700$

Normalized values for each data point:

X	Min-Max Normalized Value
400	0.000
1100	1.000
500	0.143
550	0.214
400	0.000
430	0.043
560	0.229
1000	0.857
800	0.571
600	0.286

Z-Score Normalization : Formula: $X_z = (X - \mu) / \sigma$

Where: μ (mean) = 634, σ (standard deviation) ≈ 233.14

Z-score normalized values for each data point:

X	Z-Score Normalized Value
400	-1.00
1100	2.00
500	-0.57
550	-0.36
400	-1.00

430	-0.87
560	-0.32
1000	1.57
800	0.71
600	-0.15

SUMMARY TABLE

X	Min-Max Normalized	Z-Score Normalized
400	0.000	-1.00
1100	1.000	2.00
500	0.143	-0.57
550	0.214	-0.36
400	0.000	-1.00
430	0.043	-0.87
560	0.229	-0.32
1000	0.857	1.57
800	0.571	0.71
600	0.286	-0.15

Interpretation of Results of Normalization

1. Identification of Extremes and Outliers : The normalization results help us clearly identify the extreme values and potential outliers in the dataset. Using Min-Max normalization, we observe that the value 400 appears twice and is the minimum in the dataset, resulting in a normalized value of 0. Similarly, the value 1100 is the maximum and is normalized to 1, setting the upper bound. These two values define the range for the remaining values, which fall between 0 and 1.

From the Z-Score normalization, we can interpret that values with z-scores significantly below -1 or above +1 are considered far from the mean and may indicate outliers. Specifically, the value 400 has a z-score of -1.00, indicating it is one standard deviation below the mean, while 1100 has a z-score of +2.00, suggesting it lies two standard deviations above the mean and could be a statistical outlier. Another high value, 1000, has a z-score of +1.57, which also places it in the high end of the distribution. These extreme values may warrant further investigation, as they can influence the overall data behavior and model performance.

2. Data Distribution Awareness : Normalization also helps in understanding the underlying distribution of the dataset. The calculated mean is 634, and a majority of the values fall between 500 and 600, which are relatively close to this mean. The Min-Max normalized values reveal that most entries lie below 0.3, while only a few are above 0.85, indicating that the data is right-skewed, with a longer tail on the higher side.

The Z-score values further support this observation, showing that most of the data points lie within ± 1 standard deviation of the mean, which suggests a normal-like distribution with a few high-value deviations. This distributional insight is critical for choosing appropriate statistical or machine learning models. For example, if the data were more symmetrically distributed, we might apply models assuming normality; whereas, in skewed datasets, alternative strategies might be more appropriate.

3. **Relative Positioning of Values :** The normalization techniques enable us to assess how each data point compares to others in the dataset. With Min-Max normalization, we can observe that values like 560 (normalized to 0.229) and 550 (0.214) are quite close in magnitude, indicating similar positions within the dataset's range. The Z-score normalization also reflects this, with the values having z-scores of -0.32 and -0.36, respectively, which means both are slightly below the average but not significantly different from it.

This kind of relative positioning is especially useful in classification or ranking tasks. For instance, one might categorize records as representing “low,” “average,” or “high” values based on their normalized scores. This helps in segmentation and targeted analysis, which is essential in fields like marketing or customer behavior analysis.

4. **Readiness for Further Analysis or Machine Learning :** The normalization process ensures that the data is now on a comparable scale, making it suitable for advanced analysis or machine learning applications. In particular, Min-Max normalization rescales all values to a fixed range [0, 1], which is ideal for algorithms sensitive to magnitude, such as K-Nearest Neighbors, Support Vector Machines, and neural networks. Meanwhile, Z-score normalization standardizes values based on their distance from the mean, making it suitable for algorithms that assume normally distributed data, such as linear and logistic regression.

Having data normalized eliminates the issue of some features dominating others simply because they have larger numerical ranges. This enhances the performance, accuracy, and convergence speed of many machine learning models. Therefore, normalization not only prepares the data technically but also reveals key characteristics that can guide the selection of appropriate analytical methods.

In summary, normalization of the dataset has provided valuable insights. We observed the presence of outliers and confirmed a right-skewed distribution, with most values concentrated near the mean. These findings are essential in understanding the nature of the data and in choosing suitable modeling techniques. By transforming the data into a standardized format, normalization serves as a critical step in both improving data quality and enabling effective, insightful analysis.

☞ Check Your Progress 5

Question 1: What is the purpose of Exploratory Data Analysis (EDA) in the data science process, and what tools are commonly used for it?

Answer: Exploratory Data Analysis (EDA) is a critical initial step in the data science process aimed at examining and visualizing datasets to uncover patterns, identify anomalies, test hypotheses, and check assumptions. It helps in understanding the data's structure, identifying key variables, and guiding modeling decisions. Common tools used in EDA include graphical methods such as histograms, scatter plots, box plots, and heatmaps.

Question 2: How can EDA help identify trends in sales data, and what might be a practical application of such analysis?

Answer: EDA can help identify trends in sales data by using visualizations such as scatter plots to observe patterns over time. For example, plotting monthly sales from January to December might reveal a trend of higher sales from September to January. This insight could lead to classifying the data into two time-based groups for deeper analysis. Additionally, plotting sales trends by time of day could help identify periods with higher conversion potential, aiding in strategic planning and targeted marketing.

Question 3: What is the main aim of the modeling stage in the data science process, and how does it differ from Exploratory Data Analysis (EDA)?

Answer: The main aim of the modeling stage is to build machine learning or deep learning models for predictions, object classification, or system comprehension. This stage is goal-focused, unlike EDA, which is more about discovering general trends in the data. While EDA explores and visualizes data, the modeling stage uses statistical and machine learning techniques to generate actionable outcomes.

Question 4: What are the key activities involved in the modeling stage, and why is feature selection important?

Answer: The key activities in the modeling stage include:

- Feature selection
- Model selection
- Model execution
- Model testing

Feature selection is crucial because it involves identifying variables that significantly affect the outcome. Including unnecessary or irrelevant features can reduce model accuracy. Insights from EDA and domain experts assist in choosing the right features.

Question 5: When should statistical models be preferred over machine learning models, and what is an example of such use?

Answer: Statistical models should be preferred when the objective is to

explain relationships between variables. For example, in a health department project aiming to identify high-risk pregnancies in PHCs, statistical models help understand the factors influencing risk and design interventions. They are more interpretable but less capable of handling complex patterns compared to ML models.

Question 6: What is supervised learning, and how does it differ from unsupervised and reinforcement learning?

Answer: Supervised learning uses labeled datasets where the output is known (e.g., predicting Remaining Useful Life of a cutting tool). It differs from:

- **Unsupervised learning**, which uses unlabeled data to discover patterns (e.g., clustering customer segments).
- **Reinforcement learning**, where the system learns from trial and error over time (e.g., Q-Learning).

An analogy:

- Supervised learning is like children learning from parents.
- Unsupervised learning is like learning without supervision.
- Reinforcement learning is like learning from mistakes.

Question 7: What is the role of model testing, and what metrics are used to evaluate model performance?

Answer: Model testing evaluates how well the model performs on unseen data. Key performance metrics include:

- Accuracy
- False positives and false negatives
- Mean Square Error (MSE)
- R and modified R

If the model's performance is unsatisfactory, adjustments in feature or model selection may be needed, followed by re-testing.

Question 8: Why is the choice of algorithm important in the modeling stage, and what factors influence this choice?

Answer:

The choice of algorithm is crucial because it directly affects the accuracy and interpretability of the model. Factors influencing this choice include the complexity of the problem, the type of data available (labeled or unlabeled), and the objective of the project (e.g., prediction vs. explanation). For instance, machine learning models are suitable for complex, large datasets, while statistical models are better for understanding relationships between variables.

Question 9: How do programming libraries like StatsModels and Scikit-learn support the model execution process?

Answer: StatsModels and Scikit-learn are Python libraries that simplify the process of coding and executing data science models. They provide pre-built functions and algorithms for various statistical and machine learning methods, allowing data scientists to focus on model logic and testing rather than implementing algorithms from scratch. This significantly speeds up model development and experimentation.

Question 10: What is the role of domain expertise in the feature selection process, and why is it important?

Answer: Domain expertise helps identify the most relevant variables for the model based on real-world knowledge and context. Experts can provide insights into which factors are likely to influence the outcome, ensuring that meaningful and accurate features are selected. This improves model performance and relevance, especially when EDA alone does not reveal all important variables.

Question 11: Why is the communication of findings an important stage in the data science process?

Answer: Communicating findings is essential because it enables stakeholders to understand the insights derived from the analysis. This stage supports decision-making and often leads to the deployment of models in real-world systems. Clear communication through reports, dashboards, or presentations ensures that insights are actionable and impactful.

Question 12: What are some common technical challenges faced during the integration of data science models into operational systems?

Answer: Common technical challenges include:

- Legacy software architectures that lack compatibility with modern tools
- Non-standardized data formats across departments
- Absence of consistent architectural frameworks. These issues can complicate model integration, hinder data flow, and make maintenance and scalability more difficult.

Question 13: How can organizational resistance affect the deployment of data science models, and what are some causes of this resistance?

Answer: Organizational resistance can delay or prevent the implementation of data science models. Causes include:

- Fear of job displacement
- Concerns about model reliability
- Lack of understanding of the model's benefits
- General reluctance to adopt new technologies Overcoming this resistance requires communication, training, and change management.

Question 14: What best practices can help ensure successful integration and deployment of data science models?

Answer: Key best practices include:

- Standardizing data formats and architecture
- Developing robust and well-documented APIs
- Encouraging collaboration between IT and data science teams
- Implementing models in phases (e.g., pilot projects)
- Providing training and managing change proactively These approaches facilitate smoother integration, scalability, and user adoption.

Question 15: Why are soft skills important for data scientists during the deployment phase, and which skills are especially valuable?

Answer: Soft skills are vital because they help data scientists communicate the value of their models and build trust with stakeholders. Important soft skills include:

- Communication
- Empathy
- Storytelling

These skills enable data scientists to address concerns, gain stakeholder buy-in, and support successful adoption of data-driven solutions.

☞ Check Your Progress 6

Question 1: Differentiate between Data Science and Data Analytics

Answer:

Features	Data analytics	Data science
Type of data	Deals with structured data only	Handles structured and unstructured data both
Type of intervention	Intervention is mostly done for problem solving through data analysis	Intervention is done not only to solve the problem but also to identify the problem in the first place.
Modelling and analysis	Mostly statistical analysis using statistical inferences	Predictive modelling using machine learning
Scope of analysis	Mostly focused on present time	More future looking
Ultimate Goal	To provide actionable insights that inform current decisions and improve performance in the short term.	To build intelligent systems and predictive models that can learn and adapt over time , often supporting long-term strategy.
Example – UseCase	Analyzing monthly sales data to determine which products underperformed.	Creating a machine learning model that predicts which products customers are likely to buy based on browsing history and demographics.

☞ Check Your Progress 7

1. Describe Various Data Analysis Techniques

Answer:

Data analysis can be categorized into four main types, each serving a distinct purpose in the decision-making process:

The four types of data analysis are:

- Descriptive Analysis
- Diagnostic Analysis
- Predictive Analysis
- Prescriptive Analysis

1. **Descriptive Analysis:**

Descriptive analysis aims to summarize and interpret historical data to understand what has happened in the past. It is the simplest form of data analysis, using data aggregation and visualization techniques such as charts, graphs, and tables to present trends, averages, and patterns. This type of analysis helps organizations track performance over time and report on key metrics. For example, a retail company might analyze sales data from the last quarter and find that sales increased by 15% in December compared to November. Dashboards showing total sales, customer visits, and product performance are typical examples of descriptive analysis.

2. **Diagnostic Analysis**

Diagnostic analysis goes a step deeper by investigating the causes behind trends or outcomes identified during descriptive analysis. Its purpose is to answer the question, “Why did it happen?” This analysis involves techniques like correlation analysis, root cause analysis, and data drilling to identify relationships and anomalies in the data. For instance, if a digital marketing team notices a dip in website traffic in March, they might perform diagnostic analysis and discover that a major advertising campaign ended in February, which led to reduced visibility and fewer visits.

3. **Predictive Analysis**

Predictive analysis focuses on forecasting future events or trends based on historical data. It employs statistical modeling, data mining, and machine learning algorithms to estimate the likelihood of future outcomes. This enables organizations to anticipate changes and make informed decisions. For example, a bank may use a predictive model to identify customers who are likely to default on loans by analyzing their credit history, income, and past behaviors. Such predictions allow the bank to manage risk and take preventive measures.

4. Prescriptive Analysis

Prescriptive analysis takes predictive insights further by recommending specific actions to achieve desired goals. It combines optimization, simulation, and decision modeling techniques with business rules and constraints to suggest the best course of action. For example, an airline might use prescriptive analysis to determine the best pricing strategy for flight tickets. By considering demand forecasts, competitor pricing, and booking patterns, the model recommends ticket prices that maximize revenue.

Together, these four types of data analysis form a hierarchy: descriptive analysis explains what happened, diagnostic analysis explains why it happened, predictive analysis estimates what might happen, and prescriptive analysis advises on what actions should be taken.

Check Your Progress 8

Question 1: What is the primary purpose of Exploratory Data Analysis (EDA), and how is it typically conducted?

Answer: The primary purpose of Exploratory Data Analysis (EDA) is to examine data before making any assumptions. It helps in identifying errors, understanding patterns, detecting outliers, and uncovering relationships among variables. Additionally, it aids in selecting suitable models and identifying correlations between explanatory and outcome variables. EDA can be conducted using both graphical and non-graphical methods.

Question 2: What distinguishes univariate analysis from multivariate analysis in EDA?

Answer: Univariate analysis deals with a single variable and aims to describe its characteristics and patterns without exploring relationships or causation. It uses methods like measures of central tendency, dispersion, and distribution form (non-graphical), and visual tools like stem-and-leaf plots, histograms, and box plots (graphical). In contrast, multivariate analysis involves multiple variables and examines relationships or correlations between them using both non-graphical techniques (such as cross-tabulation, correlation, and covariance) and graphical methods (like grouped bar plots, scatter plots, and heat maps).

Question 3: Why are graphical methods considered important in EDA, particularly in univariate analysis?

Answer: Graphical methods are important in EDA, especially in univariate analysis, because non-graphical methods alone often do not provide a complete picture of the data. Visualizations such as stem-and-leaf plots, histograms, and box plots help to better understand the distribution, identify outliers, and detect patterns more intuitively and effectively than numerical summaries alone.

Question 4: What are some common graphical techniques used in multivariate exploratory data analysis, and what do they illustrate?

Answer: Common graphical techniques used in multivariate exploratory data analysis include grouped bar plots, scatter plots, and heat maps. Grouped bar plots illustrate comparisons between levels of two categorical variables. Scatter plots show the relationship between two continuous variables by plotting data points on a horizontal and vertical axis, helping to reveal trends or correlations. Heat maps visually represent data values or correlations using color gradients, making it easier to detect patterns or intensity of relationships across multiple variables.

Question 5: Compare the role of data Scientist and Data Analyst in any project.

Aspect	Data Analyst	Data Scientist
Primary Role	Analyzes data to find trends (e.g., in sales or usage) to support business needs	Identifies and solves complex business problems using data; also involved in problem discovery
Interaction	Limited to analyzing data	Works closely with domain experts and top management
Decision-Making Involvement	Less involved in organizational decision-making	Actively part of the decision-making team
Required Soft Skills	Basic communication skills	Strong communication, storytelling, and management skills
Mathematical Knowledge	Algebra, probability, and statistics	Algebra, probability, statistics, and additional knowledge in machine learning
Data Types Handled	Primarily structured data	Structured and unstructured data
Tools and Technologies	SQL for querying and managing data	SQL and NoSQL; advanced tools for machine learning and big data processing
Programming Skills	Moderate programming skills	Strong programming skills
Educational Background	Undergraduate degree or short-term additional training	Typically holds a Master's or PhD in data science or related fields
Organizational Impact	Supports data-driven insights	Translates data insights into business strategies
Salary Level	Generally lower	Typically higher due to advanced skills and strategic responsibilities

Question 6; What is Crowdsourcing? Give its advantages and disadvantages.

Answer: Crowdsourcing is a powerful method of data collection that, when used responsibly, can significantly enhance the scope and quality of data science initiatives. Its success hinges on a careful balance between speed, scale, and data quality, supported by well-designed processes, validation mechanisms, and ethical practices. When implemented thoughtfully, crowdsourcing enables organizations to derive meaningful insights, make better decisions, and innovate with greater agility.

The advantages of crowdsourcing in data science are significant. It enables scalability by allowing organizations to gather large-scale data efficiently. It also brings in diversity, offering varied perspectives and data points that might not be accessible through traditional means. Furthermore, crowdsourcing often sparks innovation, as user-generated content and feedback can uncover hidden issues or inspire new features and ideas that internal teams might overlook.

However, crowdsourcing is not without its challenges. One of the most pressing concerns is data quality. Since contributors come from different backgrounds and skill levels, the data they provide can be biased, inaccurate, or inconsistent. Some users may offer incomplete or even intentionally misleading information. As a result, organizations must implement robust quality control measures to ensure the reliability of the collected data. Techniques such as proofreading, cross-validation, algorithmic filtering, and contributor reputation systems are essential to minimize errors and ensure data accuracy.

Given its strengths and limitations, crowdsourcing should be viewed as a complementary tool in data science rather than a standalone solution. It is most effective when tasks require subjective judgment, diverse perspectives, or quick, large-scale data collection that automated systems cannot achieve alone. Ethical considerations are also crucial—organizations must ensure transparency, protect user privacy, and offer value or incentives to contributors for their participation.

☞ Check Your Progress 8

1. APAAR, which stands for Automated Permanent Academic Account Registry (APAAR), is a specialized identification system designed for all students in India. APAAR ID makes it possible to personalize teaching approaches. Inspired and facilitated by APAAR ID, school management is interested in adopting a personalized teaching approach to fit each student's needs by looking at their grades, learning patterns, etc. Discuss various steps of the data science life cycle for this personalized teaching project in detail.

Answer: a detailed explanation of how each step of the **Data Science Life Cycle** can be applied to the APAAR personalized teaching project:

Step 1: Goal Identification

The goal is to develop a personalized teaching approach for students using their APAAR ID data. The objective is to analyze each student's grades,

learning patterns, and related data to tailor teaching methods that improve individual learning outcomes. This involves understanding what metrics define personalized learning success (e.g., improved grades, engagement levels) and aligning the project with school management's vision.

Step 2: Data Collection

Collect data linked to each student's APAAR ID, including historical academic records (grades, test scores), attendance, learning behavior (e.g., time spent on subjects, preferred learning resources), and demographic information. Data sources could include school databases, learning management systems, digital classroom tools, and surveys from teachers and students.

Step 3: Data Preparation

Clean the collected data by handling missing or inconsistent entries, normalizing grades to a common scale, and encoding categorical variables (such as subject preferences). Integrate data from different sources to create a unified dataset per student. Also, anonymize sensitive data to comply with privacy laws and prepare the data for exploratory analysis.

Step 4: Exploratory Data Analysis (EDA)

Perform EDA to understand trends and patterns, such as which subjects students struggle with, correlations between attendance and performance, and typical learning behaviors. Visualize data using histograms, box plots, and scatter plots to detect outliers or clusters of similar learners. This insight will help in feature selection for the model.

Step 5: Model Development

Develop predictive models to identify students' learning needs and recommend personalized teaching strategies. Use supervised learning to predict student performance or clustering algorithms to group students by learning style or difficulty level. Validate models using historical data and refine them to ensure accuracy and fairness.

Step 6: Model Deployment and Presentation

Deploy the model as a decision-support tool integrated with the school's management system. Teachers can access recommendations for personalized lesson plans or interventions through a dashboard. Present findings and model insights in an easy-to-understand format for educators and administrators, highlighting how personalized strategies can improve student outcomes.

1.14 FURTHER READINGS

1. Data Science from Scratch: First principles with Python, Joel Grus, 2nd Edition, O'REILLY, Shroff Publishers & Distributors Pvt. Ltd., 2021
2. Data Science for Beginners, Andrew Park, 1st Edition, Packt Publishing, 2020